

Harshana Lakshara Fernando

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Summary

aBackend-focused Computer Science Engineering undergraduate with a rigorous foundation in **System Design**, **Distributed Systems**, and **Cloud-Native Engineering**. Proven experience architecting high-performance, fault-tolerant applications using **Microservices**, **Serverless Architectures**, and **Event-Driven Design** with **Java (Spring Boot)**, **Go**, **AWS Lambda**, and **Kafka**. Validated through the delivery of high-throughput systems handling massive load, seeking to solve complex **concurrency** and **low-latency** challenges within high-velocity engineering teams.

Education

B.Sc. (Hons) in Engineering – Computer Science and Engineering

2022 – 2026

University of Moratuwa, Sri Lanka

G.C.E. Advanced Level (Physical Science Stream)

2019

St. John's College, Panadura, Sri Lanka

Results: 3 A's (Combined Mathematics, Physics, Chemistry)

District Rank: 17

Experience

R&D

Software Engineer — KōreroCare Platform

Mar 2026 – Present

Empathic Computing Laboratory, University of Auckland | Remote / Part-time

- Co-led backend design for the **KōreroCare** mobile-wearable sensing platform, building **Go/Gin**-based services for time-series ingestion and cross-device synchronization.
- Designed **PostgreSQL** schemas for participants, studies, devices, consent, reminders, and wearable sensor streams.
- Integrated **Keycloak OAuth2/OIDC** with **Traefik** gateway routing and Docker Compose deployment for secure internal service access.
- Developed Firebase notification and reminder flows for participant engagement and study prompts.
- Modeled an **HL7/FHIR**-aligned data schema mapping participants, devices, and wearable measurements.

Software Engineer Intern — Information Systems Associates (Pvt) Ltd

Dec 2024 – May 2025

Aviation Technology | Colombo, Sri Lanka

- **AeroOrder — Distributed PNR Generation Service**
 - Owned design and delivery of a distributed **Spring Boot/Kafka**-backed PNR generation service, **live in production**, achieving **25+ PNRs/sec** at **<100ms latency**.
 - Resolved production **Kafka consumer lag** and offset reset issues using shutdown hooks, retry handling, validation, and a Producer-Consumer refactor.
 - Led the legacy-to-microservice PNR rollout using database-driven **feature toggles** and a **strangler-pattern** migration approach.
 - Built load-testing and monitoring support using **JMeter**, **Prometheus**, and **Grafana** to validate async order/PNR flows, queue time, and service latency.
 - Maintained **>90% unit test coverage** while coordinating delivery with QA and distributed engineering teams.
- **AeroMart — Legacy Reservation System Integration**
 - Integrated **gRPC** clients and **Kafka** consumers into the **Java 8** AeroMart monolith for backward-compatible communication with AeroOrder services.
- **Engineering Practices**
 - Authored comprehensive design docs, sequence diagrams, and integration plans; facilitated knowledge transfer sessions with distributed engineering teams.

- Added **Jenkins** environment support and streamlined service logging by replacing heavyweight logging dependencies with a lightweight structured JSON logging approach.

Projects

See more projects on my portfolio [↗](#)

1. Improving Blockchain Scalability: Throughput Enhancement in ZK-Rollups

Research Project — University of Moratuwa

- Built a modular ZK-rollup benchmarking framework to evaluate Ethereum Layer-2 trade-offs across sequencing, batching, proving, data availability, and L1 settlement.
- Implemented the Rust-based Submitter service for finalized batch submission, DA-mode selection, L1 bridge interaction, transaction receipt validation, and settlement metric export.
- Added calldata, EIP-4844-style blob, and off-chain / Validium-style benchmark modes, observing approximately 70% lower DA publication cost for blob-style mode compared with calldata.
- Introduced a mock-testable **BridgeClient** abstraction, improving Submitter testability and increasing verifiable test coverage from approximately 41% to 88%.
- Built Docker-based benchmark orchestration scripts, a Poisson-style workload generator supporting 8 TPS base and 80 TPS burst workloads, and Python data tools for throughput, P50/P95/P99 latency, finalization time, gas/cost, and DA-footprint analysis.
- Produced reproducible benchmark outputs including latency CDFs, cost heatmaps, and Pareto-style trade-off comparisons for the final research evaluation.

Technologies: Rust, Solidity, Hardhat, TypeScript, Python, Docker, gRPC, SQLite/PostgreSQL, RISC0 / Mock Proving Path, Matplotlib

2. Blockchain Supply Chain Management DApp (2025 | Individual)

Architecture | Live [🌐](#)

| Github [🔗](#)

Built a full-stack decentralized application (DApp) enabling end-to-end shipment tracking and secure escrow payments on the Ethereum blockchain (Polygon Amoy Testnet).

- Developed a production-grade Solidity smart contract with a strict 3-state lifecycle (PENDING → IN_TRANSIT → DELIVERED) and on-chain auditability.
- Implemented a trustless escrow mechanism that locks funds on creation and releases payments only upon verified delivery.
- Built a responsive Next.js/React frontend with global state management and Ethers.js integration for secure wallet connections.
- Designed a CI/CD pipeline using GitHub Actions to automate contract and frontend deployment, utilizing Hardhat for local simulation and verification.

Keywords: Solidity, Hardhat, Ethers.js, Web3Modal, Next.js, React Context API, Tailwind CSS, Polygon Amoy, Smart Contracts, Escrow Payments, CI/CD

3. Governance DAO – Token-Weighted Voting DApp (2025 | Individual)



Developed a decentralized governance platform with **token-weighted voting** using Solidity smart contracts.

- Implemented **GovToken** (ERC-20) using OpenZeppelin Contracts, with faucet-based token claims, owner-controlled minting, and airdrops.
- Built **VotingDAO** smart contract enabling proposal creation, weighted voting, quorum enforcement, and post-deadline execution.
- Deployed and tested contracts on Ethereum network using **Hardhat**.

Keywords: Ethereum, Solidity, Web3.js, Hardhat, OpenZeppelin

4. Fintech & Crypto Trading Automation Lab (Ongoing | Group)

Industry-Mentored Initiative (Ongoing)

- Selected for an exclusive initiative to build intelligent automation systems for fintech and crypto markets under industrial mentorship.
- Developing production-ready trading tools and gaining deep exposure to technical analysis (TA) and algorithmic trading strategies.
- Focusing on high-performance software engineering with strict deadlines and client-side expectations.

Keywords: Fintech, Crypto Trading, Forex, Algorithmic Trading, Technical Analysis, Intelligent Automation, Industrial Mentorship, Personal Finance & Wealth Management

Professional Training, Coursework & Certificates

See all certificates [↗](#)

- **Certified Kubernetes Administrator (CKA) Training**
KodeKloud (Mumshad Mannambeth) — Completed 27h curriculum with 50+ hands-on labs. *Apr 2025*
View certificate
- **AWS Certified Solutions Architect Associate (SAA-C03) Training**
Udemy (Stephane Maarek) — Completed 27.5h comprehensive cloud architecture course. *Aug 2025*
View certificate
- **Apache Kafka Series (v3) – Learn Kafka for Beginners**
Udemy (Stephane Maarek) — Full System Design & Setup. *Mar 2025*
[Part 1] | [Part 2] | [Part 3]
- **100 Days of DevOps (Bootcamp)**
KodeKloud — Intensive hands-on training covering Linux, Python, Docker, Jenkins, Terraform & Ansible. *Nov 2025*
View certificate
- **Implementing Software Architecture for Large-Scale Systems**
Udemy *Apr 2025*
View certificate
- **Advanced OAuth Security**
Udemy *Mar 2024*
View certificate
- **Next.js: Build Scalable React Apps with Page & App Routers**
Udemy — Instructors: Anton Voroniuk, Denys Kohut, Anton Voroniuk Support *Nov 2025*
View certificate

Skills

Core	Java Spring Boot Microservices Distributed Systems System Design REST APIs
Backend	Spring Cloud Spring Security Resilience4j Node.js Express.js Gin
Messaging & APIs	Kafka RabbitMQ gRPC GraphQL WebSockets Protobuf JWT OAuth 2.0
Data	PostgreSQL MySQL Oracle DB Redis Couchbase
Cloud & DevOps	AWS Docker Kubernetes CI/CD (GitHub Actions, Jenkins)
Testing	JMeter Gatling Pytest Unit Testing
Tools	Git Gradle Postman Jira Confluence

References available upon request.